

CONTACT

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PROFILE

First-year master's student in Embedded Systems (Robotics & Drones), top of my class. I build systems end to end: VHDL logic on FPGA, real-time C/C++ firmware on microcontrollers and robotics software on Linux and ROS 2. Thorough, I validate on real hardware, with measurements.

SKILLS

Languages
C | C++ | Python | Java | VHDL | Matlab

Targets & hardware
ESP32 | ARM | Arduino | Raspberry Pi | FPGA Artix-7

Buses & protocols
UART | I²C | SPI | I²S | PWM | ADC | CAN | Bluetooth

Real time & robotics
RTOS | Embedded Linux | ROS 2 | Gazebo | LiDAR | DSP / FFT | OpenCV

Tools
Git | PlatformIO | Vivado | Simulink | Oscilloscope

LANGUAGES

French – Native
English – C1 (all courses taught in English)
German – B1

AWARDS

Top of class – EFREI Paris (2023–24 and 2025–26)
Masterdev France 2025 – national programming competition
BIA – French Aeronautics Initiation Certificate

INTERESTS

Music – guitar (15 years), bass, piano
Sport – running, cycling, tennis, skiing
3D design – Blender, Fusion 360

ANTOINE MROCZKO

EMBEDDED SYSTEMS ENGINEERING STUDENT

EMBEDDED SYSTEMS ENGINEERING INTERNSHIP (M1)

Available: internship August to October 2027 (3 months) · open to work-study in 2027–2028

EDUCATION

EFREI Paris – Panthéon-Assas University 2023 – 2028
Master of Engineering (Diplôme d'Ingénieur) – Major in Embedded Systems, Robotics & Drones
Top of class (2025–2026). International track (100 % of courses taught in English).

Concordia University, Montréal Aug. – Dec. 2025
Academic exchange semester

Lycée Saint-Dominique 2023
French Baccalauréat (Maths, Physics-Chemistry), highest honours with congratulations

Academica Dual Diploma 2019 – 2023
American High School Diploma, earned alongside the Baccalauréat

ENGINEERING PROJECTS

OctoMap Drone Explorer – Real-time 3D LiDAR mapping 2026
ROS 2 | Gazebo | 3D LiDAR | Python / NumPy | C++ (liboctomap)
A simulated drone maps a museum in 3D; the core of OctoMap rebuilt and validated:

- **ROS 2 pipeline:** LiDAR and odometry from Gazebo, real-time mapping, RViz visualisation.
- **Probabilistic map:** ray casting, octree, `.bt` export validated by liboctomap (2000/2000 voxels).
- **Sensor synchronisation:** pose interpolated at each scan: 96 % fewer false voxels.
- **Memory optimisation:** clamping tuned: 17 % less RAM with the same accuracy.
- **Quality:** pytest unit tests of the update rules and the octree, memory measurement tools.

ESP32 Audio FX – Real-time audio effects processor 2026
C++ | ESP32 | FreeRTOS | PlatformIO | I²S | SPI | ADC | Bluetooth | FFT
Complete ESP32 firmware, from the Bluetooth stream to the analog output:

- **Real-time chain:** Bluetooth A2DP audio, 15 DSP effects, I²S output to a DAC, ∼9 % CPU.
- **Interface:** TFT screen (SPI) with a 1024-point FFT, potentiometers (ADC) and buttons.
- **Dual-core FreeRTOS:** Bluetooth on core 0, high-priority audio on core 1, UI in its own task.
- **Modular code:** an `Effect` parent class, one file per effect, chain reorderable live.
- **Robustness:** click-free fades and parameter smoothing for clean live sound.

Vigibot – Autonomous industrial inspection robot (team of 5) 2026
Python | Raspberry Pi 4 | I²C | PWM | IR & US sensors | Multithreading | OpenCV
A robot that crosses 4 zones on its own, integrated, calibrated and validated on the real platform:

- **Hardware integration:** IR/US sensors, camera and PWM motors over I²C, calibrated on site.
- **Line following:** IR then camera: HSV segmentation, steering from position and direction.
- **Navigation:** ultrasonic obstacle avoidance, a labyrinth guided by arrows read by the camera.
- **Real time:** multithreaded loops, state machine between zones and YOLO object detection.
- **Supervision:** on-board web server with live video and telemetry.

Reflex game on FPGA (Arty A7) 2026
VHDL | Vivado | FPGA Artix-7 | FSM | ALU | PMOD
Synchronous hardware design in 100 % VHDL, from code to wiring on the board:

- **Logic:** state machine, reaction time measured in ms, an ALU as a random generator.
- **Outputs:** LED animations and buzzer sounds at the chosen frequency.
- **Electronics:** transistor stage on PMOD driving the 5 V buzzer, synthesis and tests in Vivado.

WORK & STUDENT ASSOCIATION EXPERIENCE

Research internship (M1) – Efrei Research Lab, Paris Nov 2026 – Apr 2027
AI-based modelling and analysis of water polo game strategy, built from player position data. Game represented as graphs; reinforcement learning (RL) to study and predict tactical choices.

Vice-President – ICE EFREI (tech association) 2026 – 2027
EFREI Paris
Technical workshops (3D printing, Arduino, Raspberry Pi) and support for students in their electronics projects.